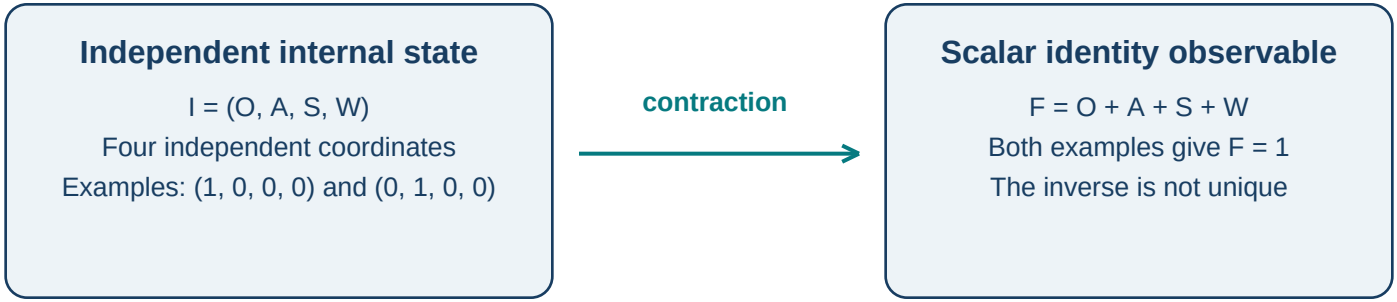

A scalar summary does not determine the full internal state.

Four internal coordinates are not a Lorentz four-vector.